

ClearHand

Multi-Agent IT Ticket Routing & Resolution

User Manual v1.0

NASSCOM Hackathon 2026 | Final Round Submission

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Chapter 1: Introduction

ClearHand is an AI-powered multi-agent system for IT service desk ticket routing and resolution. It ingests user-submitted support tickets and runs them through a pipeline of five AI agents -- Guardrail, Classifier, Router, Resolver, and Supervisor -- to produce a decision (Hand 1/2/3) with a confidence score.

The three routing outcomes are:

Hand 1 -- Self-Help: The system generates step-by-step resolution instructions for the employee to follow independently.

Hand 2 -- Team Assist: The ticket is routed to the appropriate IT department queue (Hardware, Software, Network, etc.) for an agent to resolve.

Hand 3 -- Specialist: The ticket is escalated for human specialist review (e.g., security incidents).

ClearHand provides three role-based portals:

- * Employee Portal -- Submit tickets, view resolution steps, provide feedback
- * Agent Workspace -- Manage department queue, resolve tickets, track SLA
- * Admin Console -- Dashboard with KPIs, audit log, and system monitoring

Chapter 2: Getting Started

This chapter walks you through launching the application, signing in, and selecting your workspace. ClearHand uses a demo sandbox with pre-seeded user profiles so you can explore all features without setting up a production environment.

2.1 Welcome Screen

When you first launch ClearHand (<http://localhost:8501>), you are greeted by the welcome screen. The page showcases the product branding, a dark mode toggle in the top-right corner, and three feature cards describing the core capabilities: Instant Resolution, Intelligent Routing, and Governed Escalation.

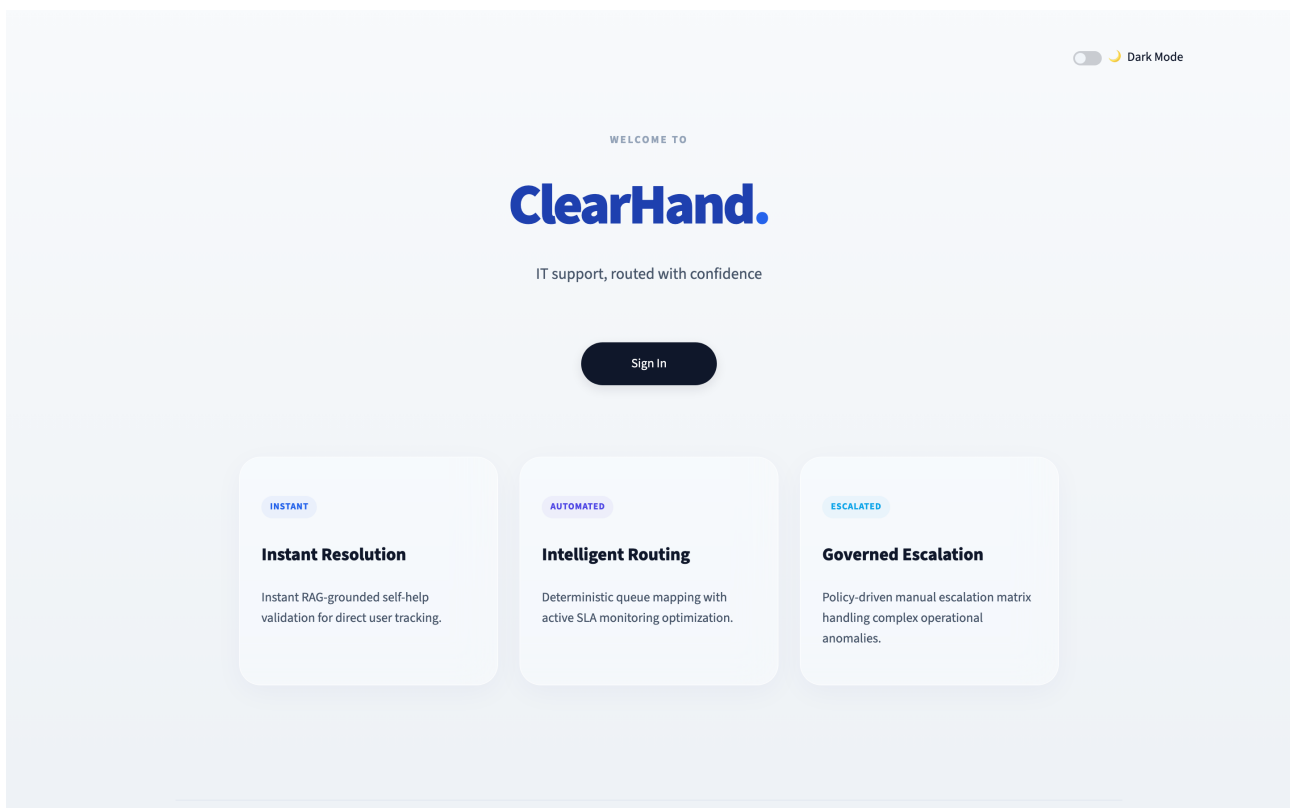


Figure 2.1: Welcome screen with product branding and feature overview

To proceed, click the 'Sign In' button centered on the screen. This takes you to the Workspace Gateway.

2.2 Workspace Gateway

The Workspace Gateway presents two operational tracks to choose from:

- * Employee Portal -- For submitting support requests and tracking resolutions.
- * Agent Workspace -- For IT agents to manage queues and service tickets.

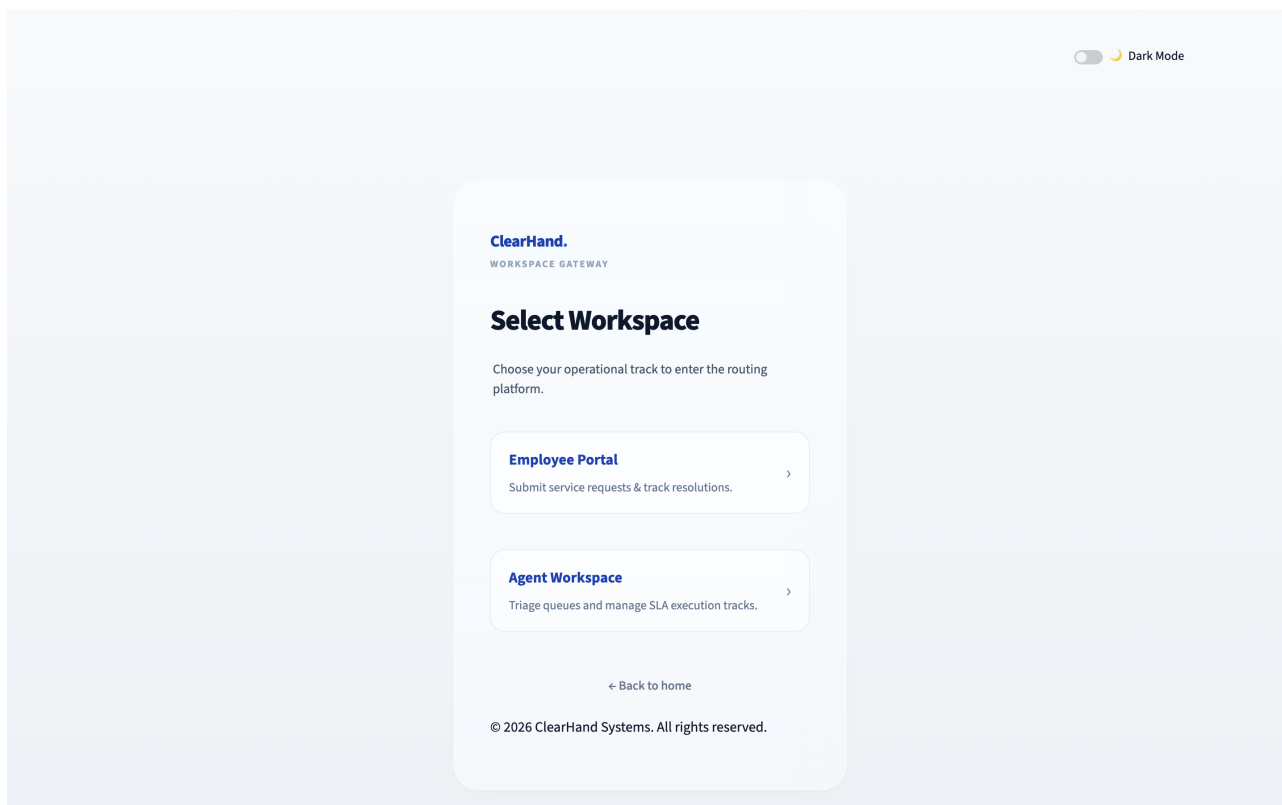


Figure 2.2: Workspace Gateway -- choose your operational track

i Tip: Hover over each tile to see a brief description. Click the tile to select that workspace.

2.3 Demo Sandbox & Login

After selecting a workspace, you enter the Demo Sandbox. This page lists pre-seeded user profiles for demonstration purposes. Each profile represents a different role and department.

Employee profiles include: Karan Joshi, Emily Reed, James Wu, Sarah Kim, Michael Brown.

Agent profiles include: Alex Chen (Hardware), Marcus Lee (Software), Priya Nair (Network), Sam Ortiz (SecOps), Jordan Kim (Identity), Riley Park (DBA), Casey Morgan (Storage), and the Global System Admin.

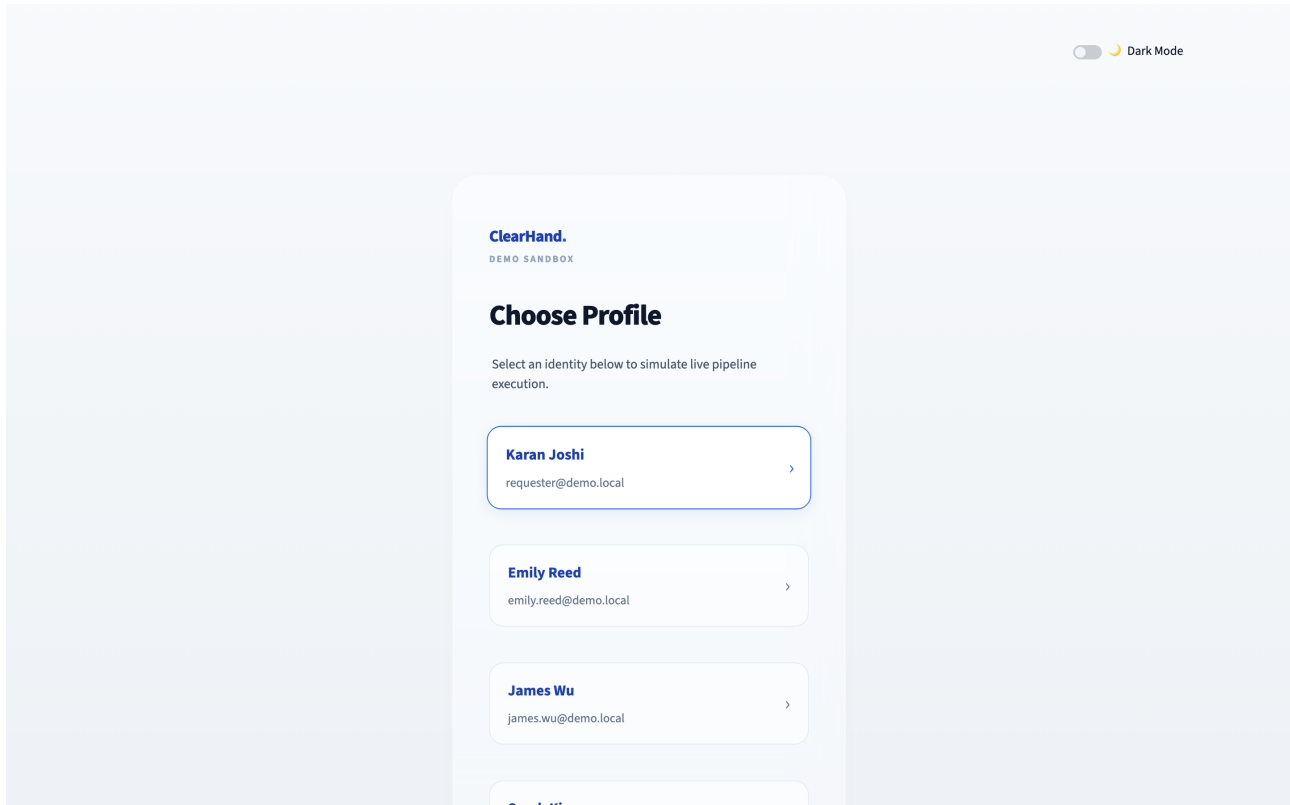


Figure 2.3: Demo Sandbox showing available employee profiles

Click on any profile to log in as that user. For this manual, we will use Karan Joshi for the Employee Portal walkthrough.

Chapter 3: Employee Portal

The Employee Portal is designed for end-users to submit IT support tickets, view resolution steps, and provide feedback. It features a Jira/ServiceNow-inspired layout with a clean, modern interface.

3.1 Home Dashboard

Upon logging in as Karan Joshi, you see the Employee Home Dashboard. The page is divided into several sections:

- * Profile Card -- Displays your name, email, and organization.
- * New Request Card -- Quick access to create a new support ticket.
- * Workspace Overview -- KPI metrics showing Open, Resolved, and Total request counts.
- * Open Requests -- A collapsible list of pending tickets.
- * Closed Requests -- A collapsible list of resolved tickets.

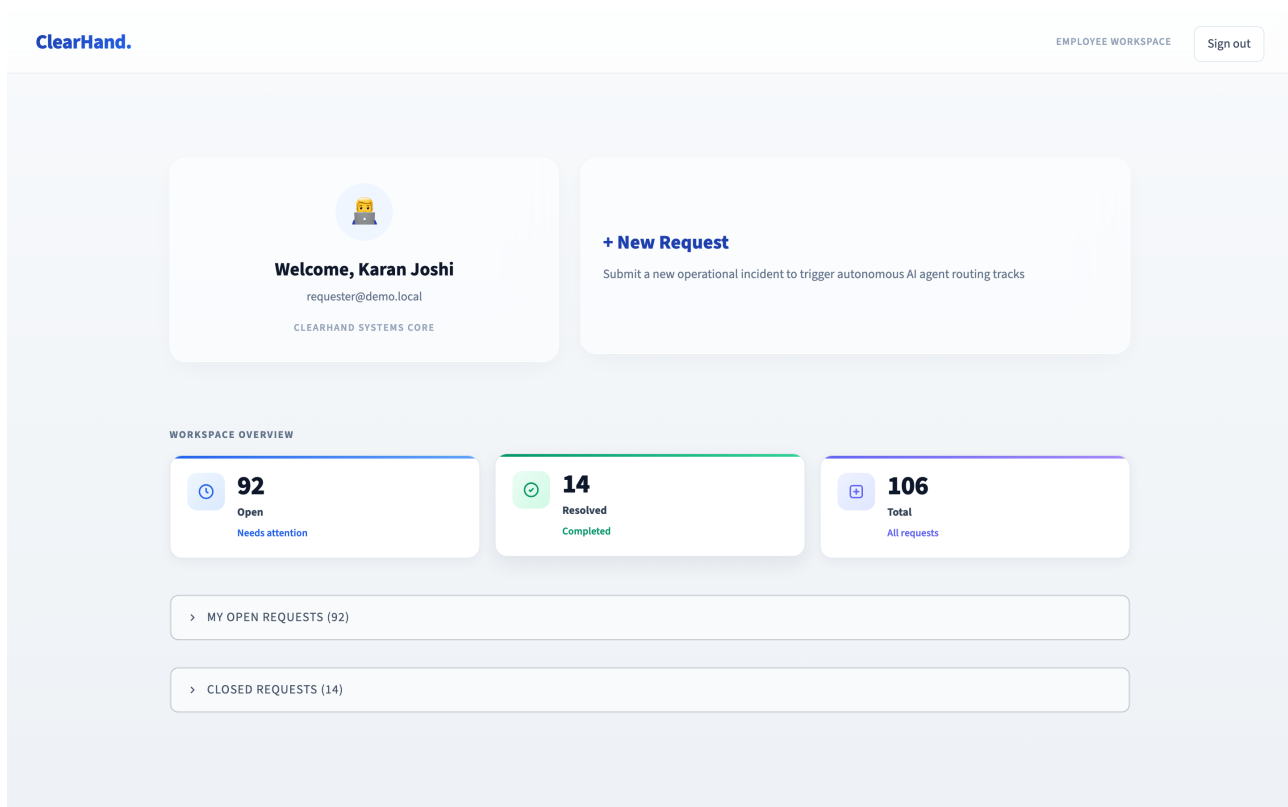


Figure 3.1: Employee Home Dashboard after login

From this dashboard, you can click '+ New Request' to begin submitting a ticket, or click 'View' on any existing ticket to see its details.

3.2 New Request Form

The New Request form collects the following information:

- * Contact Email -- Pre-filled with your email address.

- * Issue Title -- A short summary of the problem.
- * Category (optional) -- Choose from Hardware, Software, Network, Security, Access, Other, or let AI decide.
- * Urgency -- Low (P2 / 48h), Medium (P1 / 24h), or High (P0 / 4h).
- * Description -- Detailed explanation of the issue.
- * Device or Location (optional) -- Where the issue occurred.

The screenshot shows the 'ClearHand.' logo in the top left and 'EMPLOYEE WORKSPACE' with a 'Sign out' button in the top right. Below the logo is a link '← Back to workspace'. The main form is titled 'NEW REQUEST' and 'Submit a support ticket'. It includes a description field with the placeholder 'Describe the issue — our AI agents classify, route, and suggest next steps.' Below this are four sections: 'CONTACT' with a 'Contact email' field containing 'requester@demo.local'; 'ISSUE SUMMARY' with a text area containing 'e.g. Cannot connect to VPN'; 'CATEGORY (optional)' with a dropdown menu showing 'Let AI decide'; and 'URGENCY' with three radio button options: 'Low · P2 · 48h response', 'Medium · P1 · 24h response' (which is selected), and 'High · P0 · 4h response'.

Figure 3.2: New Request form with all fields visible

i Tip: For best routing accuracy, provide a clear description including error messages, when the issue started, and any troubleshooting steps you have already tried.

3.3 Self-Help Result (Hand 1)

When a ticket is classified as Self-Help (Hand 1), the system generates step-by-step resolution instructions. In this example, a password reset request is resolved with guided steps.

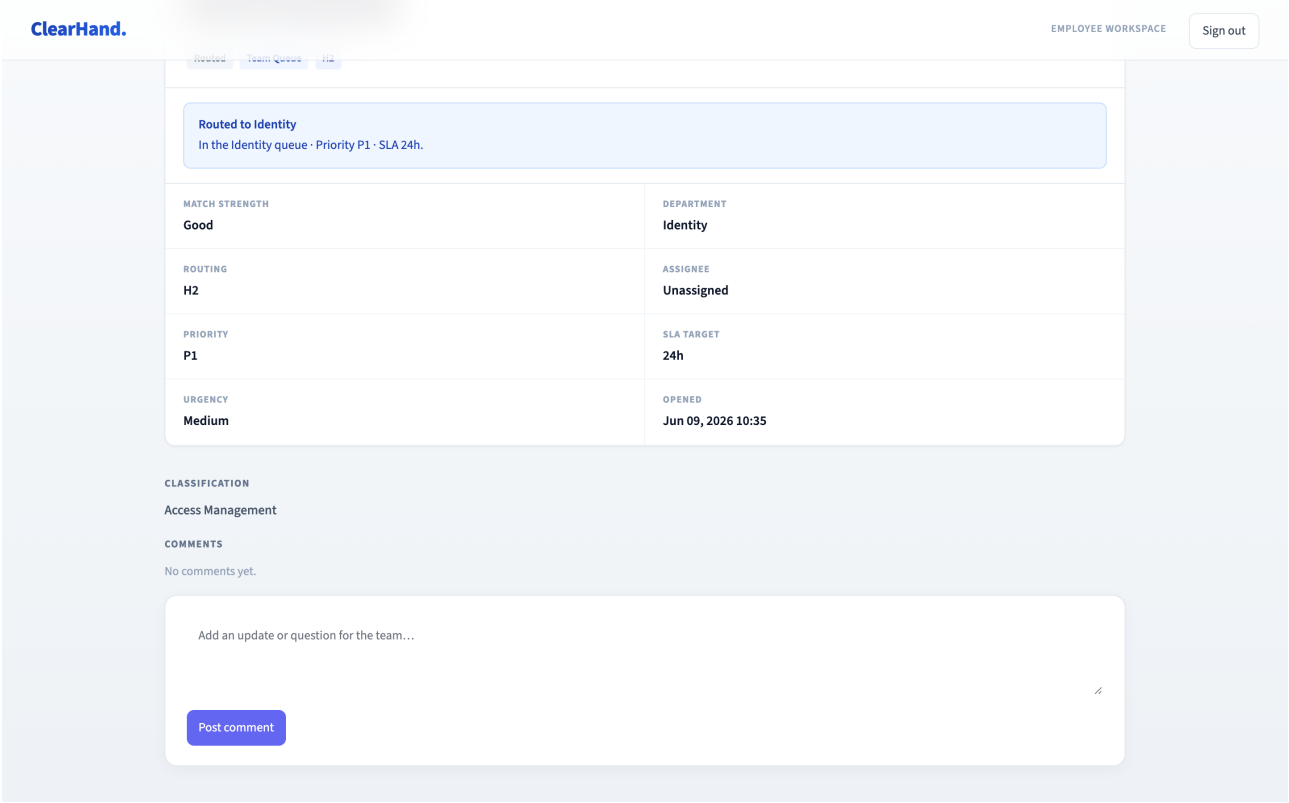


Figure 3.3: Password reset ticket resolved with Self-Help steps

After submission, the ticket detail page shows:

- * A green routing banner indicating Self-Help guidance is available.
- * The ticket metadata grid (Match Strength, Department, Routing, Priority, SLA, etc.).
- * Recommended resolution steps listed in order.
- * Two feedback buttons: 'Worked' (if the steps resolved it) or 'Did not work' (to escalate).

3.4 Team Assist Result (Hand 2)

When the system determines that human intervention is needed, the ticket is routed to the appropriate department queue. In this example, a printer hardware issue is routed to the Hardware department.

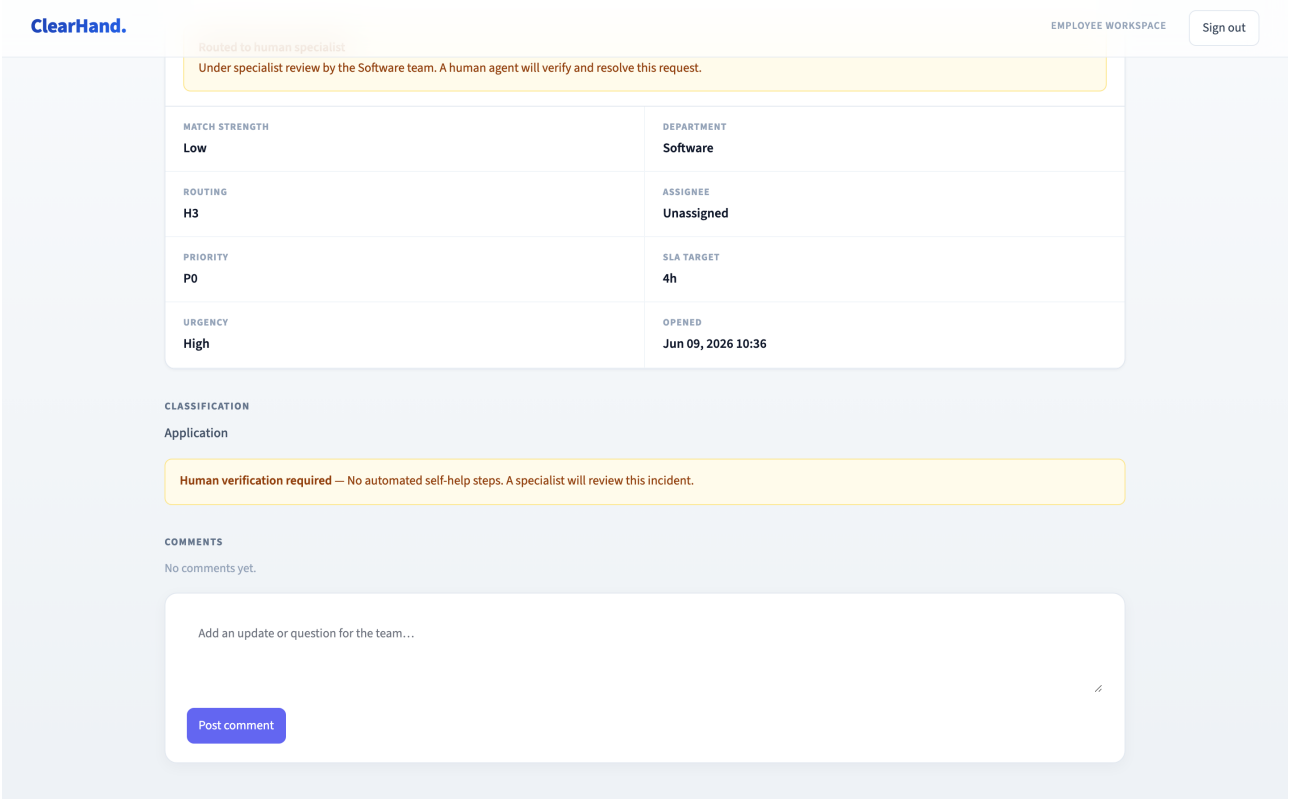


Figure 3.4: Printer issue routed to Hardware team queue

Hand 2 tickets display:

- * A blue routing banner showing the assigned department.
- * Priority and SLA information.
- * A message indicating the ticket is in the team queue awaiting an agent.
- * No automated resolution steps (the agent will handle it).

3.5 Specialist Result (Hand 3)

Security-related tickets and low-confidence classifications are automatically escalated to Hand 3 (Specialist). In this example, suspicious login activity triggers an escalation.

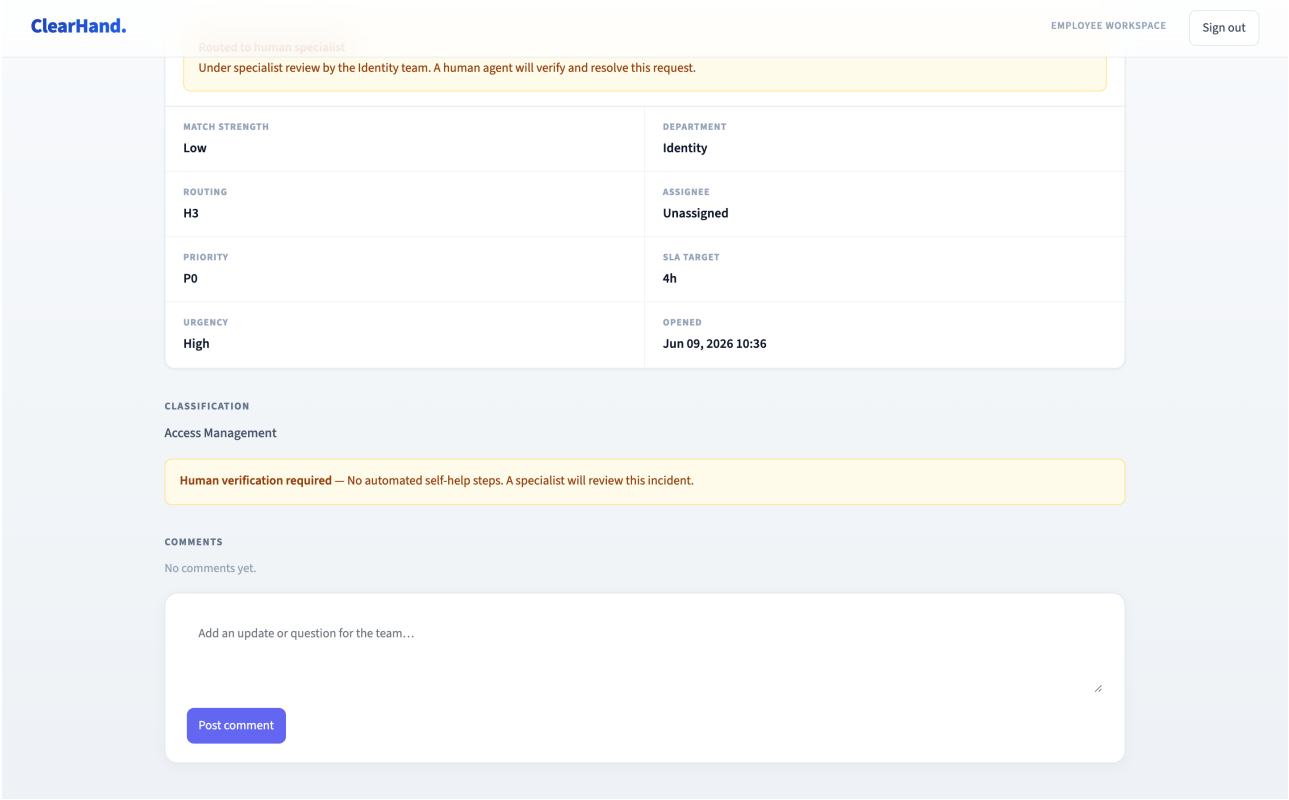


Figure 3.5: Security incident escalated to Specialist review

Hand 3 tickets show:

- * An amber warning banner: 'Human verification required'.
- * A notice that no automated self-help steps are provided.
- * Routing to the SecOps (or relevant specialist) team.
- * The ticket enters HUMAN_REVIEW status for a specialist to investigate.

3.6 Ticket Detail View

You can view any ticket's full detail by clicking 'View' from the dashboard ticket list. The detail page provides:

- * Ticket ID (INC-XXXXXXXX) and title at the top.
- * Status, routing badge, and hand classification chips.
- * A routing banner explaining the current state.
- * Metadata grid with: Match Strength (confidence), Department, Routing, Assignee, Priority, SLA Target, Urgency, and Opened date.
- * Classification category and source.
- * Resolution steps (for Hand 1 tickets).
- * Feedback buttons (for Hand 1).
- * Audit trail showing the agent pipeline trace.
- * Comment thread for communication.

ClearHand.

EMPLOYEE WORKSPACE

Sign out

Routed to human specialist

Under specialist review by the Identity team. A human agent will verify and resolve this request.

MATCH STRENGTH	DEPARTMENT
Low	Identity
ROUTING	ASSIGNEE
H3	Unassigned
PRIORITY	SLA TARGET
P0	4h
URGENCY	OPENED
High	Jun 09, 2026 10:36

CLASSIFICATION

Access Management

Human verification required — No automated self-help steps. A specialist will review this incident.

COMMENTS

No comments yet.

Add an update or question for the team...

Post comment

Figure 3.6: Full ticket detail view with metadata and audit trail

Chapter 4: Agent Workspace

The Agent Workspace is designed for IT support agents to manage their department queue, view ticket details, and resolve tickets. It provides a streamlined interface for triage and SLA management.

4.1 Agent Dashboard

Logging in as an agent (e.g., Alex Chen from Hardware) takes you to the Agent Dashboard. This page includes:

- * Profile card with agent name and department.
- * KPI metrics: In Queue count, Department name, Active agents, SLA status.
- * Team Queue section showing tickets assigned to the department.

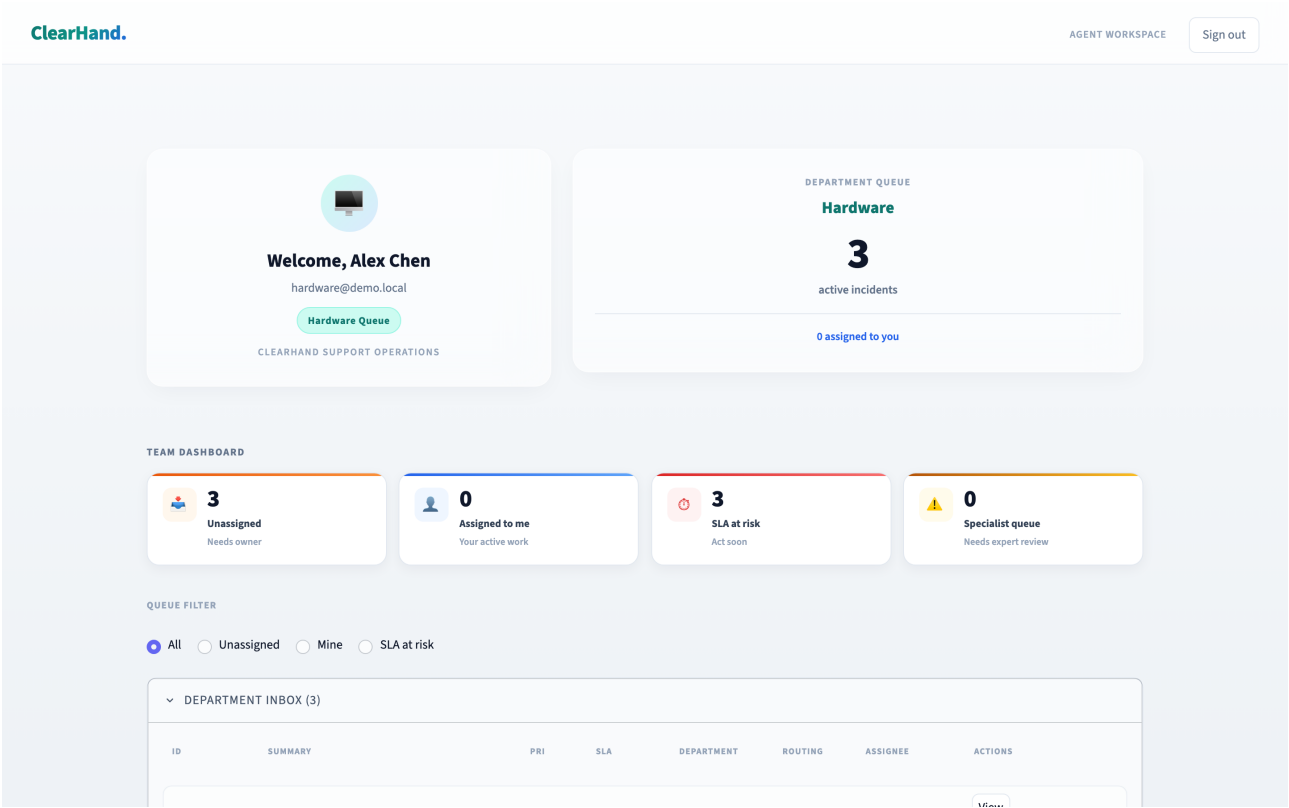


Figure 4.1: Agent Dashboard for Hardware department

i Note: Agents only see tickets routed to their specific department. A Hardware agent cannot see Software or Network queue tickets.

4.2 Team Queue

The Team Queue displays all open tickets for the department with the following columns:

- * Ticket ID (INC-XXXXXXX) -- Link to view the full detail.
- * Summary -- Brief title of the issue.

- * Department -- The assigned department.
- * Routing -- Hand classification (Self-Help, Team Assist, Specialist).
- * Assignee -- Who is working on the ticket.

Agents can click 'Open' on any ticket to view its full detail and take action.

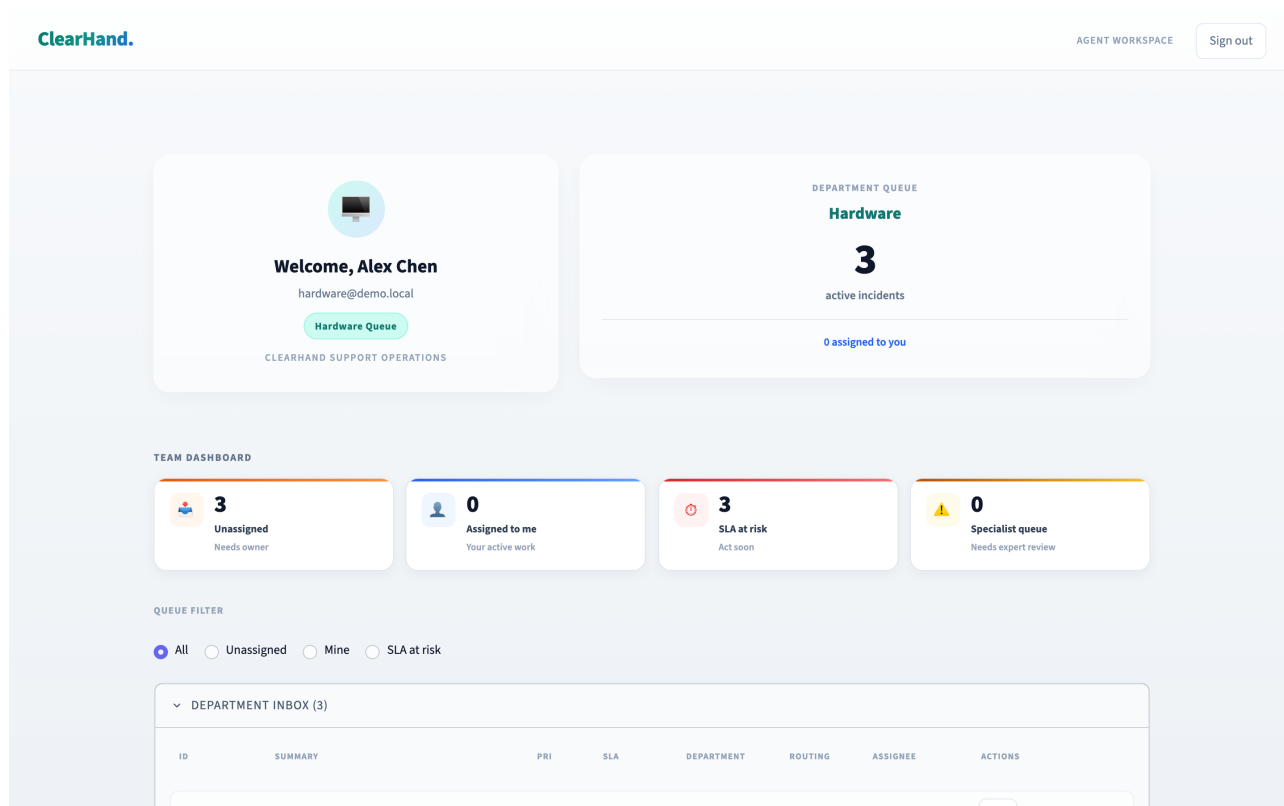


Figure 4.2: Team Queue showing open tickets for the Hardware department

From the ticket detail view, agents can:

- * View the full incident description.
- * See AI-suggested resolutions (for Hand 2 tickets).
- * Review the audit trail of agent actions.
- * Assign, release, or resolve tickets.
- * Add comments to the ticket.

Chapter 5: Admin Console

The Admin Console provides system-wide oversight with KPIs, audit logging, and compliance monitoring. Administrators can view all tickets across departments and track agent activity.

5.1 Admin Dashboard

Logging in as the Global System Admin reveals the Admin Dashboard with:

- * KPI metrics: Total Tickets processed, Audit Events count, Active Agents, Policy status.
- * Recent audit log events showing the append-only agent trace.
- * Overview of the entire system's operation.

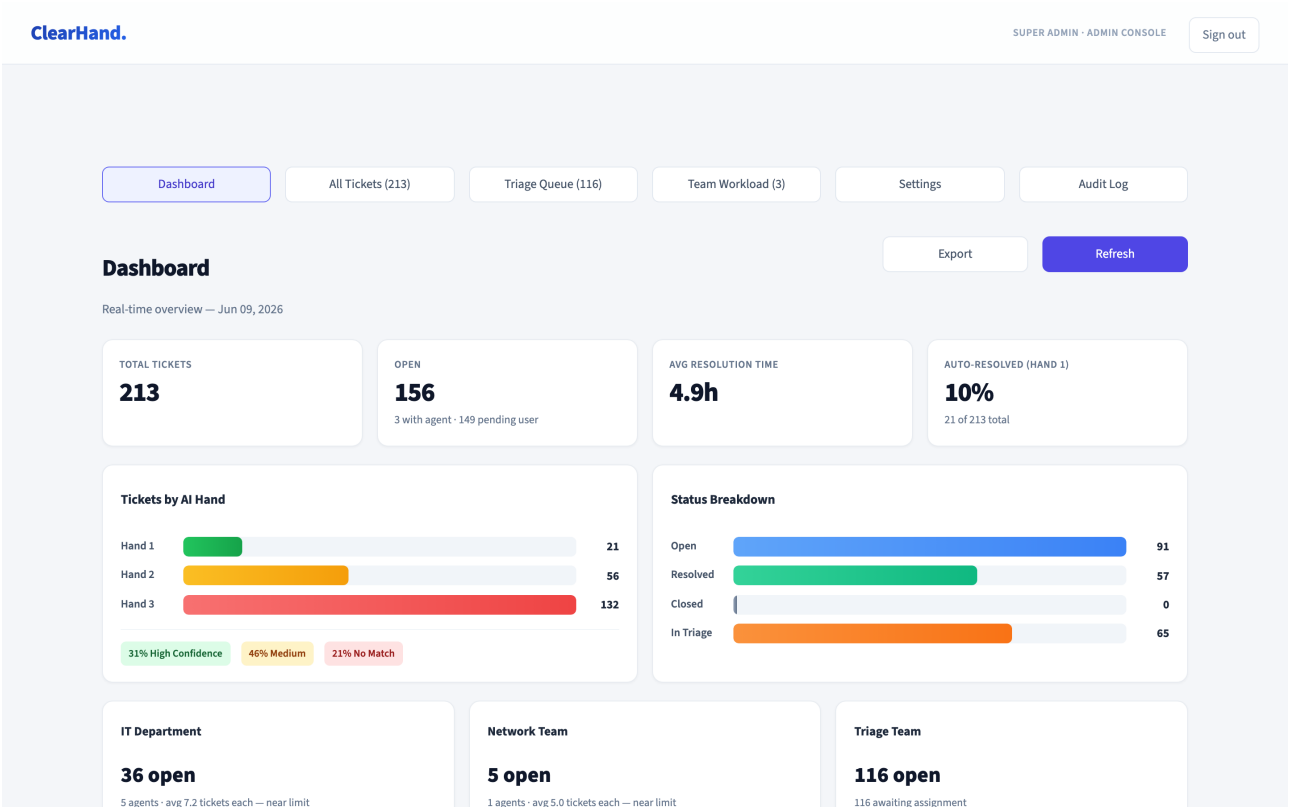


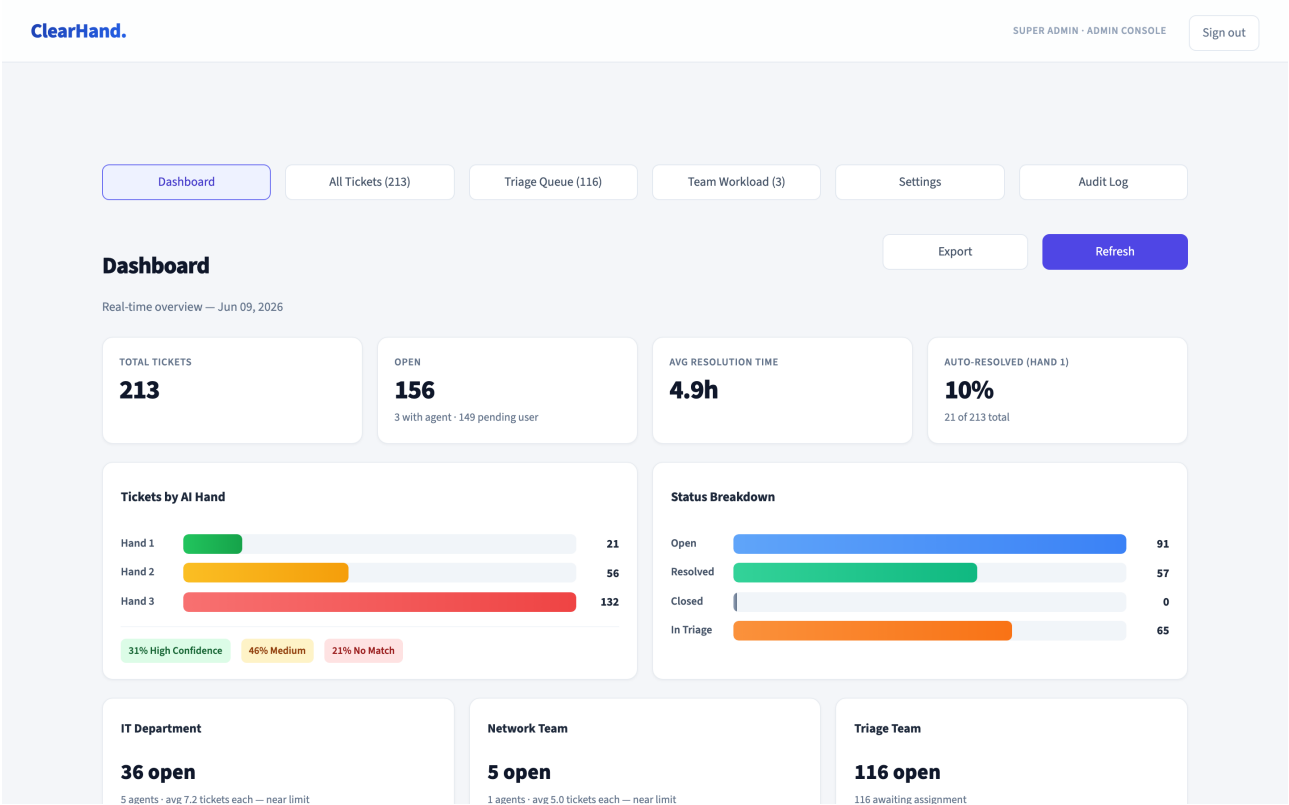
Figure 5.1: Admin Dashboard with system-wide metrics and audit events

5.2 Audit Log

The Audit Log provides a privacy-safe, append-only trace of all agent pipeline activity. Each entry includes:

- * Timestamp -- When the event occurred.
- * Ticket ID -- The affected ticket (truncated for privacy).
- * Agent -- Which agent in the pipeline (Guardrail, Classifier, Router, Resolver, Supervisor).
- * Event Type -- What action was performed.

This log is essential for compliance, troubleshooting, and auditing AI decision-making.



Chapter 6: Dark Mode

ClearHand includes a full dark mode theme for comfortable use in low-light environments. The dark mode toggle is available on the welcome screen (before login) in the top-right corner.

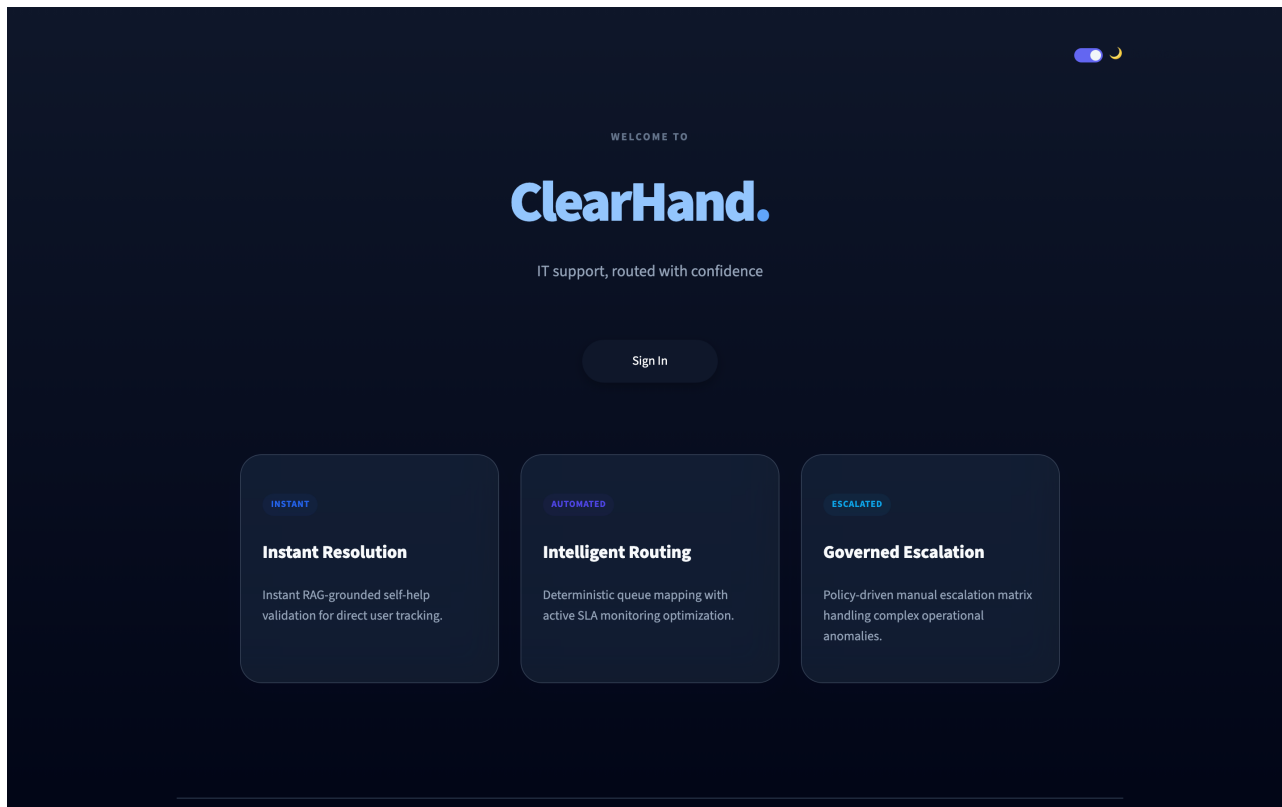


Figure 6.1: Welcome screen with Dark Mode enabled

Dark mode applies to all pages and portals, providing a consistent visual experience. The setting persists across page navigation but resets on full page refresh.

Chapter 7: Frequently Asked Questions

Q: How does ClearHand decide between Self-Help, Team Assist, and Specialist?

A: The Supervisor agent calculates a confidence score (c_total) based on RAG similarity (60%), sentiment analysis (20%), and historical success rate (20%). Security issues are always forced to Specialist (Hand 3). Tickets with low RAG grounding (<0.55 similarity) are escalated to Team Assist or Specialist.

Q: What happens when I click 'Did not work' on a Self-Help ticket?

A: The ticket is escalated from Hand 1 to Hand 3 (Specialist). The status changes to HUMAN_REVIEW, and a specialist will investigate the issue manually.

Q: How are tickets routed to the correct department?

A: The Classifier agent categorizes the ticket using Gemini LLM, RAG matching, or keyword fallback. The Router agent uses a deterministic hash-map lookup to map categories to department queues (e.g., Hardware -> Hardware, Security -> SecOps).

Q: Can I add comments to a ticket?

A: Yes. The ticket detail page includes a comment thread where employees, agents, and admins can communicate. Comments provide context for faster resolution.

Q: How does the system handle prompt injection attacks?

A: The Guardrail agent performs dual-layer defense: (1) Regex pre-check blocks override phrases before any API call, and (2) Gemini scans the ticket body for injection attempts. Failed checks force the ticket to Hand 3 / SecOps / P0.

Q: How can I reset demo data?

A: Run 'python scripts/init_db.py && python scripts/seed_users.py' to reinitialize the database with fresh demo data. Optionally run 'python scripts/seed_rag_demo_tickets.py' to seed the RAG knowledge base.